Irqued by IRQ

What's an IRQ?
Most hardware devices on your computer need attention from the processor on a fairly regular basis and the mechanism for doing this is called Interrupt Requests (IRQs), which are dedicated hardware channels over which devices can send interrupt signals to the CPU.

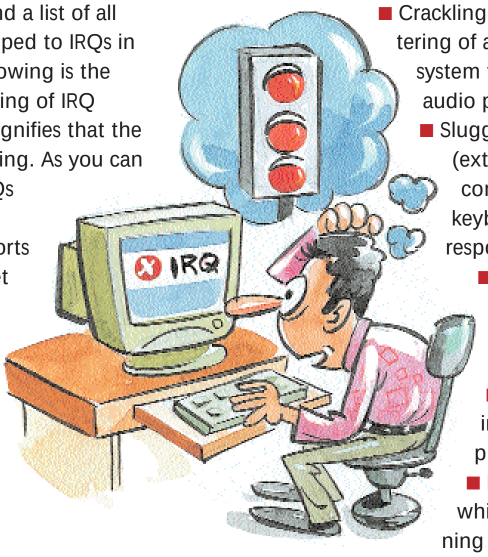
The sad part is that there are only 16 of them, and some of them are reserved for specific devices. This means devices will have to share IRQs and sharing can be a real problem if your hardware doesn't want it to share it with another.

Common IRQ assignments
Go to the System Information tool in Accessories. Proceed to ***Hardware Resources > IRQs*** and you shall find a list of all hardware that is mapped to IRQs in your system. The following is the most common mapping of IRQ assignments. Open signifies that the IRQ is free for the taking. As you can see, there are five IRQs free. Disabling the COM2 and COM1 ports from the BIOS will get you two more!

1. System Timer
2. Keyboard
3. Open
4. COM port 2
5. COM port 1
6. Open
7. Floppy
8. Parallel Port 1
9. Real Time Clock
10. Open
11. Open
12. Open
13. PS/2 Mouse
14. Math Coprocessor/FPU
15. IDE controller 1
16. IDE controller 2

Intelligently reallocating valuable IRQs to the needy will solve 90 per cent of all conflict-related problems.

Common symptoms of IRQ conflicts
- Modem won't echo commands back to the screen or simply won't work
- Hard lockups and system freezes
- Frequent data corruption while transferring data
- Crackling, skipping or stuttering of audio playback; system freezes when audio playback is initiated
- Sluggish I/O response (extremely slow input command cycles and keyboard/ mouse response on screen)
- Windows boots only in Safe mode
- Mouse freezes randomly
- Garbled or incomplete printing
- Lockups or freezes while starting or running games



Troubleshooting IRQ conflicts
- Keep the first PCI slot free as it generally shares an IRQ with the AGP slot. Graphics cards are notoriously bad at sharing resources and will usually hog the IRQ.
- Swap devices around on the motherboard. If device A and B are conflicting then change device A with device C. This is time consuming, since you have to uninstall/reinstall drivers.
- If all else fails, the only way out is to remove a device that you don't use or could do without. Alternatively, exchange it for a different model with different resource requirements.
- Windows 98 SE handles IRQ sharing much better than Windows 98. Also Windows 2000 and Windows XP handle resource allocation completely differently from previous Windows versions and probably have the best chance of getting your combination of hardware to work. The advantage they have is that they simply refuse to install on detecting conflicting devices.
- It is always a good idea to free up as many unused resources in the system as possible before you start trying to resolve a conflict. You should be able to disable allocation of resources such as IRQs to COM ports, USB ports and Printer ports (LPT) in the BIOS if you do not use them.
- Refer to your motherboard manual for IRQ assignments for specific PCI slots. You will find that certain PCI slots will share IRQs irrespective of the component installed. Avoid installing problem devices in these slots.

PRINTERS

Printer shows 'out of memory'

My LaserJet printer has suddenly started giving me an error saying, "out of memory." What could be the problem?

There could be many reasons for receiving this error. Each printer has a distinct set of printer drivers which instruct on how the task is to be handled. This particular driver may have been changed or swapped or the settings might have been changed.

Some of these devices require the user to specify the amount of RAM available. Hence, check the settings for the printer. Go to ***Start > Settings > Printers***, right-click on Printer Driver and go to Properties. There you will be able to see this option. The setting here should match the rated setting of the printer. If this is not a problem, then it may be that the printer spools all tasks to the hard disk before printing any task and over a period of time your hard disk may have become clogged with all this data, leaving you with no space to spool the task. This can be fixed by clearing some space from your hard drive.

There's another remote possibility: the memory chips might have become loose or perhaps been damaged. In that case, you need to contact a technician to get the problem solved.

Prints garbled text

My printer prints garbled text. What could be the problem?

There could be two reasons for such behaviour: either the drivers you installed are not correct or are corrupt, or there could be a fault in the printer head such as worn out pins or badly rotating pin cushion rollers. If the problem doesn't disappear even after reinstalling the drivers, call a technician for help.

Error writing to LPT1

Whenever I try to print documents, I get a message saying, "Error writing to LPT1". What do I do?

This is a very common error and there are quite a few ways to fix it. First, make sure that the printer is online and that there is sufficient paper loaded in the trays. Then clear the printer's